

Statistical Analysis of the Relationship Between MIC and BPC values for tobramycin among *Pseudomonas aeruginosa* isolates: Do isolates with higher tobramycin MIC values also tend to have higher BPC values?

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Introduction

Pseudomonas aeruginosa can grow either as free-living planktonic cells or as a biofilm.These growth conditions may respond differently to antibiotics.The minimum inhibitory concentration (MIC) is the lowest antibiotic concentration that prevents visible planktonic growth, whereas the biofilm prevention concentration (BPC) is the lowest concentration that prevents biofilm formation under the tested conditions [1]. In this report,following question addressed ***'Do isolates with higher tobramycin MIC values also tend to have higher BPC values?'***To answer this question, analysed made from the publicly available MIC and BPC measurements, the GitHub repository associated with the original study.

Data sample

Two data files, *MIC_IMC.csv* and *BPC_IMC.csv*, were used in the analysis [2].Both files contained tobramycin measurements for *P. aeruginosa* isolates. The relevant variables were:

- isolate ID: unique identifier for each biological isolate;
- MIC: concentration that prevents visible planktonic growth, measured in µg/mL;
- BPC: concentration that prevents biofilm formation, measured in µg/mL.

Each original file contained 60 rows because every one of the 30 biological isolates had two technical replicates. These replicates were checked for consistency and collapsed into one record per isolate to avoid incorrectly doubling the sample size.The MIC and BPC datasets were then matched using isolate identifiers, resulting in 30 correctly paired MIC–BPC observations. The concentration values followed discrete two-fold steps and included many tied observations.Therefore, the association between MIC and BPC was measured using Spearman's rank correlation.Sampling uncertainty was assessed using 10,000 paired bootstrap resamples. MIC and BPC were resampled together to preserve the biological pairing within each isolate.

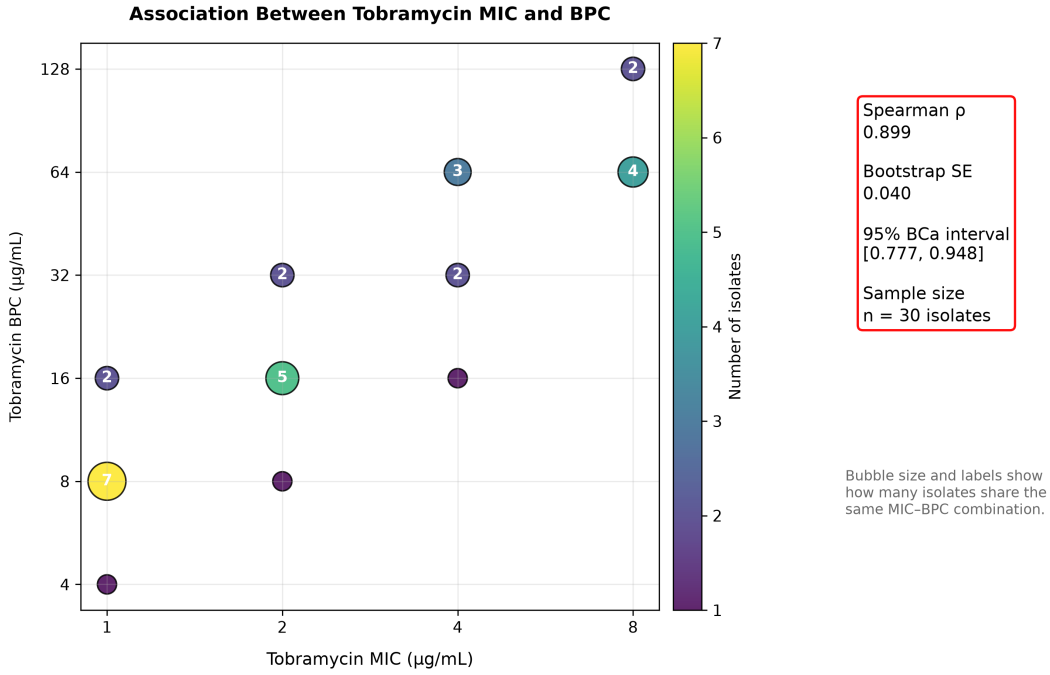


Figure 1: ***Association between tobramycin MIC and BPC in 30 P. aeruginosa* isolates.**

Results and discussion

Figure 1 shows an overall upward pattern between MIC and BPC.Isolates with higher MIC values generally also had higher BPC values.The median MIC was 2 µg/mL, while the median BPC was 16 µg/mL.The main descriptive and inferential results are summarized in Table 1.

Table 1: Descriptive and inferential results.

Summary measure	**Result**
Number of matched isolates	30
Median MIC	2 μg/mL
Median BPC	16 μg/mL
Spearman correlation	$\rho_s = 0.899$
Bootstrap standard error	0.040
95% BCa interval	0.777–0.948

The strength of the relationship was measured using Spearman's rank correlation: The Spearman correlation was calculated as

$$\rho_s = \mathrm{cor}(R_X, R_Y) = 0.899$$

where Rx and Ry represent the ranks of the MIC and BPC values.A correlation of 0.899 indicates a strong positive association. However, this does not mean that MIC and BPC are equal or that a higher MIC causes a higher BPC.Some isolates had identical MIC and BPC combinations.Ordinary points would therefore overlap and hide observations.For this reason, bubble size and the number inside each bubble show how many isolates share the same concentration pair. A paired nonparametric bootstrap method was used to assess the uncertainty in the correlation estimate.In each bootstrap iteration, 30 MIC–BPC pairs were resampled, and the Spearman correlation was recalculated.This process was repeated 10,000 times.Since the MIC and BPC values belonged to the same isolate, they were resampled in a paired manner.Additionally, a 95 percantage of bias-corrected and accelerated (BCa) confidence interval was calculated to account for potential bias and asymmetry in the bootstrap distribution [3]. The variation between the resulting correlations was summarized by the bootstrap standard error:

$$SE_{\mathrm{boot}} = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\rho_s^{*(b)} - \overline{\rho_s^*}\right)^2} = 0.040$$

Here, $B = 10{,}000$ is the number of bootstrap samples, $\rho_s^{*(b)}$ is the Spearman correlation obtained from bootstrap sample b, and $\overline{\rho_s^*}$ is the mean of all bootstrap correlations. The relatively small standard error shows that the estimated correlation did not vary substantially across the bootstrap samples.The 95 percentage of BCa confidence interval ranged from 0.777 to 0.948.

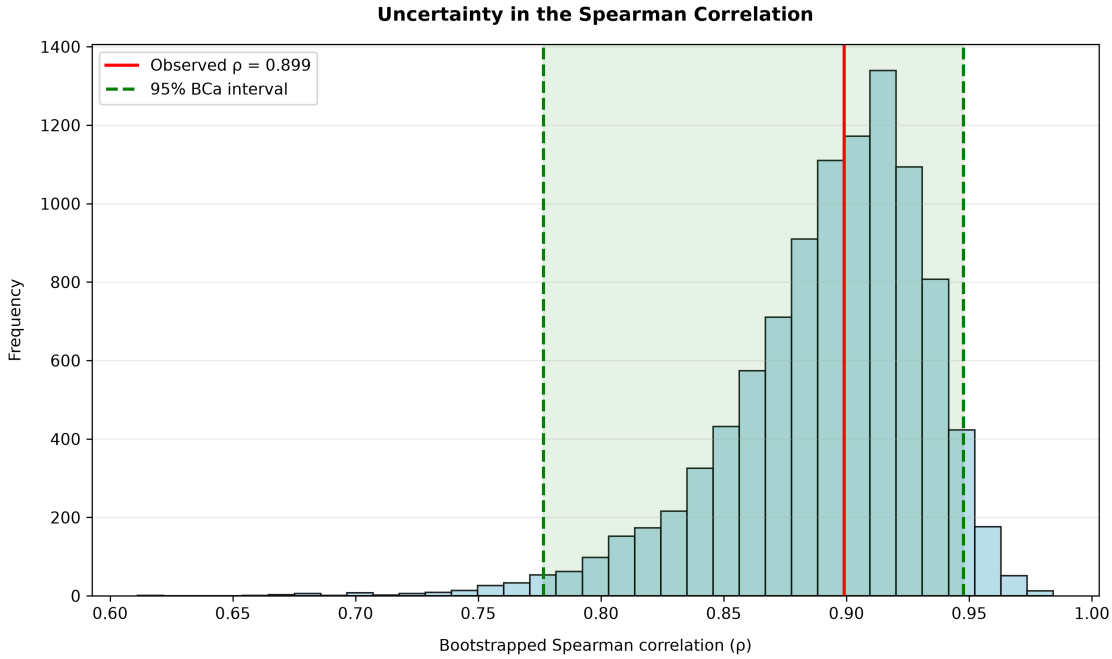


Figure 2: ***Distribution of Spearman correlations from 10,000 paired bootstrap resamples. The solid red line represents the observed correlation, while the dashed green lines represent the 95% BCa confidence interval.***

Figure 2 shows that most bootstrap correlations were concentrated around the observed value of 0.899.Importantly, the lower confidence limit of 0.777 still represents a strong positive association.This indicates that the conclusion was not dependent on a single unusually high correlation estimate.

Overall, isolates with higher tobramycin MIC values tended to have higher BPC values. The paired bootstrap analysis suggests that this relationship was strong and stable within the analyzed dataset. However, the result represents an association rather than causation and should not automatically be generalised beyond the 30 isolates included in the study.

References

[1] Vergauwe, F., De Waele, G., Sass, A., et al. (2025). Harnessing machine learning to predict antibiotic susceptibility in *Pseudomonas aeruginosa* biofilms. *npj Biofilms and Microbiomes*, 11, 205. Available at: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12603021/` DOI: `https://doi.org/10.1038/s41522-025-00833-4`

[2] De Waele, G. (2025). *biofilm-amr: Code and data supporting machine learning for antimicrobial resistance prediction in Pseudomonas aeruginosa biofilms*. GitHub. Available at: `https://github.com/gdewael/biofilm-amr/tree/main/data/isolates`

[3] Efron, B. (1987). Better bootstrap confidence intervals. *Journal of the American Statistical Association*, 82(397), 171–185. Available at: `https://doi.org/10.1080/01621459.1987.10478410`